Factual Self-Awareness in Language Models: Representation, Robustness, and Scaling

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Abstract

Factual incorrectness in generated content is one of the primary concerns in ubiquitous deployment of large language models (LLMs). Prior findings suggest LLMs can (sometimes) detect factual incorrectness in their generated content (i.e., fact-checking post-generation). In this work, we provide evidence supporting the presence of LLMs' internal compass that dictate the correctness of factual recall at the time of generation. We demonstrate that for a given subject entity and a relation, LLMs internally encode linear features in the Transformer's residual stream that dictate whether it will be able to recall the correct attribute (that forms a valid entity-relation-attribute triplet). This self-awareness signal is robust to minor formatting variations. We investigate the effects of context perturbation via different example selection strategies. Scaling experiments across model sizes and training dynamics highlight that self-awareness emerges rapidly during training and peaks in intermediate layers. These findings uncover intrinsic self-monitoring capabilities within LLMs, contributing to their interpretability and reliability.[1]

1 Introduction

With the recent advents in the ability of Large Language Models (LLMs), security concerns associated with LLM-based AI agents have increased in direct proportion (Huang et al., 2024). A lion's share of the security concerns about everyday LLM usage comes from their tendency to spit out made up facts— a tendency mainly addressed under the broad description of *hallucination*, bearing an insinuation of *forgetfulness* of the models (Huang et al., 2025). Recent research seeking to address hallucination has posed the question of transparency: is there any generalizable signal that can inform the presence (or absence) of certain knowledge in the internal representation of the model?

Prior research in this direction follows two distinct lines. Kadavath et al. (2022) showed that language models (LMs) can (most of the time) fact-check their output. Multiple subsequent studies (Li et al., 2023a; Azaria and Mitchell, 2023; Burns et al., 2023) have investigated and found that the LM encodes a notion of truth (and false) as linear directions within its representations, and these directions causally elicit the internal fact-checking. This is similar to the broader line of research embodied in the self-reflection paradigm (Madaan et al., 2023; Pan et al., 2023): let the model generate first and ask it to check itself. Recent literature has challenged the paradigm itself in problems such as reasoning and planning (Stechly et al., 2025).

Another line of investigation reveals that language models (LMs) can demarcate between known and unknown entities (Ferrando et al., 2024). This demarcation, defined as the LMs ability (or inability) to recall at least three attributes about the candidate entity correctly, is also linearly represented within

[1]https://github.com/UKPLab/arxiv2025-self-awareness

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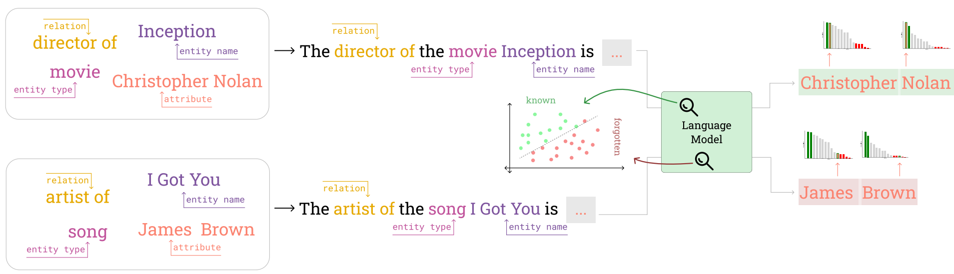

Figure 1: Given an input comprising entity type, entity name, and relation, we obtain the model's token-level prediction probabilities for the attribute. Tokens are labeled `known` if their gold label appears in the top-k predictions, and `forgotten` if in the bottom-l. A sample is labeled `known` if it contains more `known` than `forgotten` tokens, and vice versa. Probabilities are visualized with color-coded bars: green (top-k), red (bottom-l), and gray (others). For example, "Christopher Nolan" falls in the top-k, labeling the sample as `known`, whereas "James Brown" appears in the bottom-l, labeling it as `forgotten`. Final token residuals are linearly probed to detect factual self-awareness.

the residual stream of the Transformer model and instruction-tuned models re-purpose these linear features for refusal of questions related to unknown entities. As opposed to the 'truth direction'-style literature that analyzes the truthfulness of LMs on checking their own outputs, Ferrando et al. (2024) impose the notion of transparency at the time of generation.

However, factual hallucination is not associated only with novel entities. An LM may incorrectly recall a specific attribute of an entity while accurately recalling another (e.g., in the dataset we prepared, 1865 out of 3669 unique entities are neither completely known nor completely forgotten; Gemma-2 2B (Team et al., 2024) can recall at least one attribute about them correctly while erring on at least one attribute). Such a hallucination is intuitively harder to detect, compared to known-unknown entities in line with Ferrando et al. (2024): an unknown entity (potentially unseen in the training data) is associated with a hitherto unseen lexical combination of a few successive tokens, whereas an unknown (or forgotten) factual association needs to be understood in a much deeper connection between entities and relationships. In this work, we show that LMs encode meta-knowledge (i.e., ability or inability to correctly recall) about the fine-grained factual relationships as linear directions (see Figure 1). These directions are activated *before* generating the correct (or incorrect) factual recall, as opposed to the truth directions that are triggered after the generation is complete and the model is asked to check the generation. This internal separability between correct and incorrect generations, which we denote as *known* and *forgotten* factual associations, is surprisingly robust to context perturbation. We further investigate the effects of training and parameter counts on the factual self-awareness of the model; while a minimum model size is required to start encoding the signal, we find that it does so very early in training.

Contribution. Towards investigating factual self-awareness of LMs at generation time, we construct a factual recall dataset. We investigate using Gemma-2 models (2B and 9B) (Team et al., 2024), and the Pythia scaling suite (Biderman et al., 2023), and propose a model-dependent annotation of known-forgotten facts based on logit distribution. We show that LMs construct linear subspaces within internal representations that can demarcate between an upcoming correct/incorrect recall (as opposed to faithfulness in post-hoc checking of correct/incorrect facts). The effects of context and prompt formatting on the formation of these linear subspaces of self-awareness are investigated. Finally, we demonstrate the appearance and improvement of factual self-awareness across two directions of LM scaling: amount of training for next token prediction and model parameters.

2 Background and Related Work

In this section, we review existing literature dissecting language models' (LMs) self-awareness and provide background on linear probes and sparse autoencoders (SAEs), including their prior use in investigating truthfulness and self-awareness in LMs.

Self-awareness of LMs. Prior efforts to make LMs transparent about their mistakes have primarily focused on mitigating hallucinations. A common approach is to ask LMs unanswerable questions and test their ability to refrain (Yin et al., 2023; Bajpai et al., 2024). However, these questions are unanswerable not due to unknown or forgotten factual associations. Betley et al. (2025) similarly studied behavioral self-awareness in LMs. By contrast, our work specifically targets factual self-awareness. Prior work in this area follows the 'self-reflexion' (Madaan et al., 2023; Pan et al., 2023) paradigm: ask the model to reflect on its generation and assess its correctness. Kadavath et al. (2022) supported LM self-awareness by demonstrating the success of such reflection on both multiple-choice and open-ended questions. They also introduced fine-tuning strategies to calibrate model-generated scores of knowledge uncertainty. Kapoor et al. (2024) extended this 'teaching to be self-aware' paradigm via calibration tuning so that model-generated logits better reflect internal uncertainty.

Truthfulness and hallucination in internal representation. Several works (Li et al., 2023a; Azaria and Mitchell, 2023; Burns et al., 2023) have investigated how truthfulness is represented in model internals (i.e., intermediate layer outputs) in post-generation self-checking setups similar to Kadavath et al. (2022). The viability of self-consistency (Zhang et al., 2024; Wang et al., 2023) as a proxy for self-awareness has also been explored. Chen et al. (2024) demonstrate an implicit assumption of self-awareness via self-consistency: among a population of diverse generations, certain spectral patterns of internal states signal hallucination. These methods are primarily designed for reasoning-based tasks, where multiple generations can lead to the same answer, in contrast to immediate factual recall. Ji et al. (2024) analyzed training data to study unseen queries and found that LMs linearly represent seen vs. unseen queries in hidden states. In a related direction, Ferrando et al. (2024) examined how LMs internally demarcate known from unknown entities. They identified linearly encoded features that trigger when the model is queried about an unknown entity—features repurposed in chat-tuned models to elicit refusal. A key limitation, however, lies in defining knowledge at the entity level: when an LM fabricates factual associations, it may invent attributes for a known entity.

Linear and sparse probing. Linear probes are arguably the simplest lens for examining high-dimensional neural representations—given a labeled dataset of an expected behavior, a linear classifier is trained and tested on neural representations to detect whether the behavior is encoded. Sparse autoencoders (SAEs) Bricken et al. (2023); Huben et al. (2023), by contrast, have recently gained popularity for uncovering interpretable decompositions of model latent representations without supervised data. Both approaches align with the linear representation hypothesis Park et al. (2023); Mikolov et al. (2013), which posits that interpretable features—such as sentiment or truthfulness—are embedded as linear directions within the representation space, and that model representations consist of sparse linear combinations of these directions Li et al. (2023b); Zou et al. (2023). While SAEs eliminate the need for supervision, they introduce the challenge of costly training: they must be trained on large volumes of data (and corresponding activations) to avoid data bias (Kissane et al., 2024; Sharkey et al., 2025). SAE dimensionality strongly influences the contextual scale of interpretation (Bussmann et al., 2025). In this work, we first show the empirical similarity between linear probes and SAEs in locating factual self-awareness, then adopt linear probes for scalability.

3 Experimental Setup

Dataset. We construct a factual recall dataset covering four categories: football players, movies, cities, and songs, following the approach of Ferrando et al. (2024). We start with 1,000 entities for each of the following four categories: football player, movie, city, and song, limiting ourselves to a maximum of 10 relationships per entity. For each entity we scrape associated features from Wikidata Vrandečić and Krötzsch (2023). Subsequently, we manually construct templates from the triplets[2] (*entity type*, *entity name*, *relation*) to generate statements and predict the corresponding *attribute*, as illustrated in Figure 1.

Due to the web-scale pretraining data used in training the Transformer-based LMs (as well as the non-availability of training data in case of most open-weight models), it is non-trivial to demarcate which factual associations are known (or unknown) to the LM. We use a proxy definition for the same, using the logit distribution of the LM itself: if a model is able to signal that it can (or cannot)

[2]Factual associations are typically represented as triplets of the form (*entity name*, *relation*, *attribute*); we additionally use entity type to avoid ambiguities arising from shared naming across entities, e.g., computer scientist *Michael Jordan* vs sportsman *Michael Jordan*.